The Metro Is the Market:

Why Local Airbnb Regulations May Not Lower Rents

Daryl Larsen*

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Abstract

Restricting short-term rentals can return housing to residents within a municipality without substantially increasing residential housing availability across its metropolitan market. Visitors displaced from regulated locations may stay elsewhere in the same metro, inducing landlords there to move housing into short-term use. I develop a spatial equilibrium model that makes this offset explicit. With minimal assumptions, local regulation reallocates short-term rentals without changing residential rents. Short-term rental prices rise and visitor utility falls. Evidence presented for New York City's Local Law 18 shows higher short-term rental prices, an outward shift in short-term rental activity, and no sustained decline in residential rents in the reported comparisons, consistent with the model predictions.

1 Introduction

A municipality that restricts short-term rentals may substantially reduce listings within its borders yet produce little improvement in housing affordability. The direct mechanism behind such regulation is straightforward: housing previously occupied by visitors becomes available to long-term residents. The equilibrium effect depends on what happens elsewhere. If visitors can obtain accommodation in another part of the same metropolitan area, their relocation increases demand for short-term housing there. Landlords outside the regulated jurisdiction may respond by withdrawing housing from residential use. A local increase in residential availability can therefore be offset by a decrease elsewhere in the market.

*Sauder School of Business, University of British Columbia

The central argument of this paper is that the metropolitan area is the relevant market for housing. A visitor's destination can be a metropolitan centre even when the visitor's accommodation is in a different municipality. Residents likewise compare locations within a broader housing and commuting market. Municipal boundaries determine where a rule applies; they need not determine the set of locations over which households substitute. When the rest of the metro can absorb displaced activity, the policy may change where housing is used by tourists much more than it changes the total amount of housing available to residents. This distinction concerns the geographic scope of market adjustment, rather than the effectiveness of enforcement within the regulated area.

New York City's Local Law 18 provides the setting for examining this mechanism. Enforcement began in September 2023. The policy required registration of short-term rentals and prohibited platforms from processing unregistered bookings, sharply reducing Airbnb listings within the city. The relevant alternatives for visitors, however, extend beyond the five boroughs. Accommodation elsewhere in the metropolitan area can still provide access to the same employment, cultural, and tourist centre. The empirical question is therefore whether the local contraction reduced metropolitan demand for tourist housing or primarily changed its location.

I formalize this distinction in a monocentric spatial equilibrium model with residents, visitors, and absentee landlords. Residents and visitors travel to a common centre and require fixed quantities of housing. Visitors have a steeper bid-rent gradient, while landlords differ in the cost of supplying short-term accommodation. Regulation limits eligibility for short-term use. In the benchmark, the metropolitan populations of residents and visitors are fixed, and eligible locations can accommodate displaced visitors after the restriction. Total housing requirements then remain unchanged. With full occupancy and a residential fringe tied to a fixed outside land return, the residential rent schedule is unchanged as well. Visitors instead pay more for accommodation and relocate toward eligible locations farther from the centre.

The exact rent invariance result is a benchmark with explicit conditions. It does not follow from spatial substitution alone. If the restriction reduces metropolitan demand for tourist housing, overwhelms accommodation capacity elsewhere, changes housing production or neighbourhood amenities, or alters resident migration, the result can change. The broader implication is an attenuation mechanism: greater scope for adjustment within the metropolitan market reduces the connection between a local decline in short-term rentals and an aggregate increase in residential

availability. Large local changes in listings need not imply large metropolitan affordability gains.

The model yields four predictions that organize the analysis. First, residential rents and renter utility are unchanged in the fixed-demand benchmark. Second, a binding restriction raises short-term bid rents and lowers visitor utility. Third, a uniform tightening shifts the distribution of short-term use outward under a stated shape restriction on hosting costs. Fourth, a local tightening induces additional short-term use in eligible areas beyond the policy boundary and, in the stylized geography, a discontinuity in short-term supply at that boundary. The last prediction directly connects the geographic scope of regulation to the capacity of the rest of the metro to absorb displaced demand.

Residential rent comparisons are considered both within the New York metropolitan area and against synthetic comparisons outside it. The results are consistent with the model's predictions, though in some cases are imprecisely estimated. Long-term rents see no change while short-term rental prices are higher in the post-period price-distance profile, and activity shifts toward peripheral locations. These findings are consistent with the model's mechanism, but the available estimates do not establish that all displaced activity remained in the metro or that the causal rent effect is exactly zero.

The distinction between municipal and metropolitan effects also changes the interpretation of empirical comparisons. Areas outside NYC are potential recipients of displaced activity, so they cannot automatically be treated as unaffected controls. A comparison of NYC with the rest of its metro measures differential adjustment within an integrated market. An estimate of the aggregate metropolitan effect requires a counterfactual outside that market. Evidence on quantities, prices, and geography must be interpreted jointly to distinguish reallocation from an aggregate reduction in tourist housing demand.

Section 2 positions the argument in the literature. Section 3 develops the model and its four predictions. Section 4 explains how absorption capacity and metropolitan visitor demand govern departures from the benchmark. Section 5 presents the empirical setting, designs, and evidence. Section 6 discusses the implications for the geographic scope of housing policy.

2 Related Literature

The first relevant literature studies how short-term rentals affect housing allocation and affordability. Evidence on rents and house prices includes Barron, Kung, and Proserpio (2021), Garcia-Lopez et al. (2020), and Franco and Santos (2021). Calder-Wang (2022) studies the welfare and distributional effects of short-term rentals in an integrated housing model. The mechanism here is compatible with tourist demand raising housing costs. Its contribution is to distinguish a reduction in tourist housing demand throughout a market from a restriction that primarily reallocates that demand within the market. The effect of local regulation need not be the reverse of the effect of introducing short-term rentals across the entire market.

A second literature examines restrictions on short-term rentals. Koster, Van Ommeren, and Volkhausen (2021) studies municipal restrictions in Los Angeles County. Related work includes studies of regulation by Bibler et al. (2021, 2025), and of spatial spillovers by van Holm (2020) and Valentin (2021). The contribution here is to connect displacement across regulatory boundaries to joint predictions for residential rents, short-term rental prices, and the location of short-term use. A change in listings inside a jurisdiction is an intermediate outcome; its affordability effect depends on the response outside that jurisdiction as well.

The theoretical framework builds on the monocentric city tradition of Alonso (1964), Mills (1967), and Muth (1969). Related applications incorporating short-term renters include Sheppard and Udell (2016) and Wrede (2022). I use a deliberately simple model to isolate the accounting and spatial equilibrium conditions under which local housing conversions are offset elsewhere. The paper's market-boundary argument also relates to the literature on spatial responses to place-based policy, including Glaeser and Gottlieb (2008) and Fajgelbaum and Gaubert (2020). It does not imply that all place-based policies are ineffective: the result depends on the relevant substitution margins and the market failure a policy addresses.

3 A Model of Short-Term Rental Regulation

The model separates the economic market from the jurisdiction imposing the restriction. The developed urban area represents the metropolitan market. A municipality occupies only part of that area. The model is intentionally kept as simple as possible to highlight the key mechanisms

and keep the analysis transparent.

3.1 Households, travel, and bid rents

Locations are indexed by distance $d \geq 0$ from a common centre containing jobs and visitor destinations. Housing and land have a fixed conversion ratio, normalized to one. There are $N_L > 0$ resident households and $N_S > 0$ visitor parties present during a common model period. Each type $i \in \{L, S\}$ requires a fixed amount $\ell_i > 0$ of housing. Write $M_i = N_i \ell_i$ for its total housing requirement. In particular, N_S is the stock of visitor parties requiring accommodation during the period, rather than an annual count of arrivals.

Type i has budget w_i, travel cost $t_i > 0$ per unit of distance, pays location rent $R_i(d)$,and continuous, strictly increasing utility $u_i(x)$ from other consumption. A household chooses location and consumption subject to

$$x + R_i(d)\ell_i + t_i d = w_i.$$

Spatial indifference within each type implies common equilibrium consumption $x_i = u_i^{-1}(\bar{u}_i)$ and the bid-rent schedule

$$R_i(d) = \frac{w_i - x_i}{\ell_i} - \beta_i d, \qquad \beta_i = \frac{t_i}{\ell_i}. \tag{1}$$

Assume $\beta_S > \beta_L > 0$. A smaller housing requirement, a larger travel cost, or both can make visitors' bid-rent gradient steeper. Define

$$\Delta(d) = R_S(d) - R_L(d) = A - \kappa d, \qquad \kappa = \beta_S - \beta_L > 0. \tag{2}$$

The intercept A is an equilibrium object. The two bid schedules cross at $d^* = A/\kappa$ when $A > 0$.

3.2 Landlords and regulation

Absentee landlords own a continuum of parcels and choose to rent to short-term visitors or long-term residents. Short-term use entails a net operating cost ε per unit of land. Costs at each location follow the same continuous cumulative distribution F, with $F(0) = 0$, support $[0, \bar{\varepsilon}]$ or $[0, \infty)$, and strict increase on its support.

A fraction $\alpha(d) \in [0, 1]$ of land at d is eligible for short-term use. Eligibility is independent

of hosting costs.[1] An eligible landlord compares residential rent $R_L(d)$, net short-term revenue $R_S(d) - \varepsilon$, and the outside land return $R_A \geq 0$. An ineligible landlord chooses between residential and outside use. In a fully occupied region with $R_L(d) \geq R_A$, the short-term land share is

$$\Gamma(d) = \alpha(d)F(A - \kappa d). \tag{3}$$

Eligibility and actual use are distinct: some eligible landlords prefer residential tenants. This formulation corresponds to independent eligibility across parcels, rather than a quota assigned to the lowest-cost hosts.

I consider two policy experiments. Under uniform regulation, $\alpha(d) = \alpha > 0$ throughout the metropolitan market. Under local regulation, a certain area of land is exempt from short-term use regulations. The latter experiment isolates the mismatch between the policy boundary and the market boundary. A radial cutoff is a stylization of NYC's geography, rather than a literal representation of its borders.

3.3 Equilibrium and the capacity to absorb displaced demand

Assumption 1 (Maintained metropolitan equilibrium)**.** *Resident and visitor housing requirements M_L, M_S are fixed. Before and after the policy, the developed area is fully occupied, has a residential fringe with a fixed outside land return R_A, and admits an interior crossing $0 < d^* < D$, where D is the metropolitan edge. Budgets permit strictly positive other consumption. The short-term clearing function defined below is strictly increasing over its positive relevant range.*

An equilibrium consists of consumption levels, bid rents, a land allocation, and an urban edge satisfying household choice, landlord optimization, and separate market-clearing conditions:

$$M_S = \int_0^D \Gamma(d)\,\mathrm{d}d, \tag{4}$$

$$M_L = \int_0^D [1 - \Gamma(d)]\,\mathrm{d}d, \tag{5}$$

$$R_L(D) = R_A. \tag{6}$$

[1] Have received critical comments here. Need additional justification or alternative specification.

The first two equations allocate the housing required by each group. Their sum is the central accounting identity.

Lemma 1 (Metropolitan housing requirements)**.** *Under assumption 1,*

$$D = M_L + M_S, \qquad R_L(d) = R_A + \beta_L(D - d). \tag{7}$$

Proof. Adding eqs. (4) and (5) gives $M_L + M_S = \int_0^D 1 \, \mathrm{d}d = D$. The boundary condition and the residential bid-rent slope then give the second expression. □

The remaining equilibrium object is A, determined by

$$G(A; \alpha(\cdot)) \equiv \int_0^D \alpha(d) F(A - \kappa d) \, \mathrm{d}d = M_S. \tag{8}$$

Given this root,

$$R_S(d) = R_A + \beta_L D + A - \beta_S d, \tag{9}$$

$$x_L = w_L - R_A \ell_L - t_L D, \tag{10}$$

$$x_S = w_S - \ell_S(R_A + \beta_L D + A). \tag{11}$$

The absorption condition has substantive content. To give sufficient conditions for this equilibrium, suppose $x_L > 0$ and define

$$A_{\max} = \min\left\{\kappa D, \frac{w_S}{\ell_S} - R_A - \beta_L D\right\}.$$

Require $A_{\max} > 0$ and

$$G(A_{\max}; \alpha(\cdot)) > M_S > 0 \tag{12}$$

under each policy being compared. Continuity, $G(0) = 0$, and the maintained strict increase imply a unique $A \in (0, A_{\max})$. Consumption is positive and the crossing occurs before the metropolitan edge. Residential bids exceed the outside return for $d < D$, while both types' bids are below that return beyond D. Thus the allocation is consistent with full occupancy and a residential fringe.

For uniform positive eligibility the clearing function is strictly increasing on its positive interior

range. The central-restriction policy with positive peripheral eligibility has the same property once the short-term support reaches eligible peripheral land. Arbitrary eligibility schedules need not have this property, so I do not claim uniqueness for every spatial policy.

For a complete ban on $[0, b]$ and peripheral eligibility $\alpha_o > 0$, eq. (12) becomes

$$M_S < \alpha_o \int_b^D F(A_{\max} - \kappa d)\, \mathrm{d}d. \tag{13}$$

The weaker condition $M_S < \alpha_o(D - b)$ is necessary but not sufficient: available land must also be profitable to convert at feasible visitor bids. A sufficiently extensive restriction can violate these conditions. In that event, the fixed-population interior equilibrium characterized here no longer applies. The assumption that the rest of the metro can absorb the change is therefore a condition to evaluate, rather than a consequence of defining a large market.

3.4 Four predictions

Proposition 1 (Residential rents and renter utility)**.** *For any two policies satisfying assumption 1 with the same primitives, residential rents at every location and resident renter utility are unchanged.*

Proof. Fixed M_L and M_S imply fixed D by lemma 1. Equations (7) and (10) then fix $R_L(d)$ and x_L, hence $u_L(x_L)$. □

The local accounting makes the offset explicit. Let M_i^c and M_i^o denote housing allocated to type i inside and outside the regulated municipality. For a fixed municipal area and unchanged metropolitan requirements,

$$\Delta M_L^c = -\Delta M_S^c, \qquad \Delta M_S^o = -\Delta M_S^c, \qquad \Delta M_L^o = -\Delta M_L^c, \tag{14}$$

where Δ here denotes a policy-induced change. Housing returned to residents within the municipality is matched by housing shifted to visitors elsewhere. This is an equilibrium reallocation across uses, not a claim that the same residents or visitors remain at their original addresses. The rent result additionally uses the fringe and fixed housing requirements; the accounting identity alone is not a general theorem about housing prices.

Proposition 2 (Short-term rents and visitor utility)**.** *Let $\alpha_1(d) \leq \alpha_0(d)$, with a strict reduction on a positive-measure set of locations with positive initial short-term use. If both policies satisfy assumption 1, then $A_1 > A_0$ and*

$$R_{S,1}(d) - R_{S,0}(d) = A_1 - A_0 > 0.$$

Visitor consumption falls by $\ell_S(A_1 - A_0)$ and visitor utility declines.

Proof. At the original intercept, the restriction gives $G(A_0; \alpha_1) < M_S$. Restoring clearing with a strictly increasing G requires $A_1 > A_0$. Equations (9) and (11) establish the rent and consumption changes. □

The increase is a common *additive* shift in bid rent per unit of housing per model period. Observed short-term rents exist only where short-term rentals operate. Under uniform eligibility, when differentiation is valid,

$$\frac{\mathrm{d}A}{\mathrm{d}\alpha} = -\frac{M_S/\alpha}{\alpha \int_0^D f(A - \kappa d)\,\mathrm{d}d} < 0, \tag{15}$$

provided the denominator is positive. For illustration, uniform costs on $[0, \bar{\varepsilon}]$ and an unsaturated intercept $A \leq \bar{\varepsilon}$ yield

$$M_S = \frac{\alpha A^2}{2\bar{\varepsilon}\kappa}, \qquad A = \left(\frac{2\bar{\varepsilon}\kappa M_S}{\alpha}\right)^{1/2},$$

subject to the same feasibility and interiority conditions.

Proposition 3 (Outward reallocation under uniform regulation)**.** *Suppose additionally that F is log-concave on its positive support. A uniform reduction from α_0 to $\alpha_1 < \alpha_0$ weakly raises $\Gamma_1(d)/\Gamma_1(0)$ relative to $\Gamma_0(d)/\Gamma_0(0)$ at fixed distances in the old short-term support. The spatial distribution of short-term use shifts outward:*

$$H_1(d) \leq H_0(d), \qquad H_j(d) = \frac{1}{M_S} \int_0^{\min\{d,D\}} \Gamma_j(s)\,\mathrm{d}s.$$

Proof. Log-concavity makes the reversed hazard f/F non-increasing. Thus $F(A - \kappa d)/F(A)$ is weakly increasing in A at fixed d, establishing the normalized-share comparison. On the old positive

support,

$$\frac{\Gamma_1(d)}{\Gamma_0(d)} = \frac{\alpha_1}{\alpha_0}\frac{F(A_1 - \kappa d)}{F(A_0 - \kappa d)}$$

is non-decreasing in d. This implies weak single crossing of the allocation change, allowing an interval of equality. Since $A_1 > A_0$, there is positive entry on (d_0^*, d_1^*). Equal aggregate short-term use requires offsetting losses closer to the centre. Every partial integral of $\Gamma_1 - \Gamma_0$ is therefore non-positive, giving the stated distributional ordering. □

This proposition gives a precise interpretation of a flatter supply profile: a higher relative share at peripheral locations and an outward distributional shift. A reduction in the absolute slope requires the additional condition that f be non-increasing. Empirical comparisons should use fixed geographic bins and distinguish shares from levels. In particular, a peripheral location can gain share when total listings fall without gaining any listings in absolute terms.

Proposition 4 (Local regulation and absorption beyond the border)**.** *Start with uniform eligibility $\bar{\alpha} > 0$ and reduce eligibility to $\alpha_c < \bar{\alpha}$ on $[0, b]$, where $0 < b < d_0^*$. Maintain assumption 1 under both policies. Then:*

1. *Short-term use weakly increases outside the municipality, strictly wherever $F(A_1 - \kappa d) > F(A_0 - \kappa d)$. New entry occurs on (d_0^*, d_1^*).*

2. *Short-term use falls throughout the municipality if $\alpha_c < \bar{\alpha}F(A_0 - \kappa b)$, and vanishes there under a complete ban.*

3. *The post-policy short-term land share jumps at the border by*

$$\Gamma_1(b^+) - \Gamma_1(b^-) = (\bar{\alpha} - \alpha_c)F(A_1 - \kappa b) > 0.$$

4. *Wherever short-term rentals survive, their bid rents rise by the common amount $A_1 - A_0$.*

Proof. The restriction raises A by proposition 2. Outside the municipality eligibility is unchanged, so monotonicity of F implies weakly higher short-term use. The expansion of d^* gives entry. Inside, initial use is at least $\bar{\alpha}F(A_0 - \kappa b)$, while post-policy use is at most α_c, proving the sufficient condition for a decline. Continuity of $F(A_1 - \kappa d)$ gives the border jump. The rent result is proposition 2. □

The local-policy result does not require log-concavity. At already saturated peripheral locations, use need not increase further, even though rents rise. Under the fixed-demand benchmark, the aggregate increase outside the municipality exactly absorbs its loss of short-term use. In data, the extent of this absorption is an object to estimate. Neither a map of relative shares nor a positive peripheral growth rate alone establishes that it is complete.

4 When Can a Local Restriction Improve Affordability?

The metropolitan perspective identifies both the offsetting mechanism and its limits. The fixed-demand benchmark allows a strong local contraction without an aggregate change in visitor housing requirements. A departure from that benchmark requires specifying which aggregate demand or supply margin adjusts.

4.1 Metropolitan demand for tourist housing

Hold N_L, housing requirements per household, and the other primitives fixed, but allow N_S to vary while retaining the residential-fringe regime. At a location occupied by residents in both equilibria,

$$\mathrm{d}R_L(d) = \beta_L \ell_S \, \mathrm{d}N_S. \tag{16}$$

More generally, if effective tourist housing use M_S changes while residential housing requirements remain fixed, the same accounting gives $\mathrm{d}R_L(d) = \beta_L \, \mathrm{d}M_S$. Within this model, the relevant change for affordability is a reduction in metropolitan tourist housing requirements. A decline in the regulated municipality's listings is insufficient if it is offset elsewhere.

One explicit demand extension lets $N_S = \mathcal{D}(C)$, where

$$C = \ell_S R_S(d) + t_S d = \ell_S (R_A + \beta_L D + A)$$

is the location-invariant generalized accommodation and travel cost of a stay. Define the metropolitan demand elasticity $\varepsilon_D = \mathrm{d}\ln N_S / \mathrm{d}\ln C$. For a differentiable uniform policy change and positive residential rents,

$$\frac{\mathrm{d}\ln R_L(d)}{\mathrm{d}\ln \alpha} = \frac{\beta_L \ell_S N_S}{R_L(d)} \, \varepsilon_D \, \frac{\mathrm{d}\ln C}{\mathrm{d}\ln \alpha}. \tag{17}$$

The final term is the total equilibrium response with endogenous visitor demand; it cannot be replaced by the fixed-population derivative in eq. (15). Nor is a listing-level own-price elasticity the relevant metropolitan elasticity. Visitors who leave one listing or municipality may still demand housing within the same metropolitan market.

If a housing-share representation is useful, the appropriate share is

$$s = \frac{N_S \ell_S}{N_L \ell_L + N_S \ell_S}, \qquad \frac{\mathrm{d}s}{\mathrm{d}\ln N_S} = s(1-s).$$

The equivalent chain-rule decomposition is

$$\frac{\mathrm{d}\ln R_L(d)}{\mathrm{d}\ln \alpha} = \frac{\partial \ln R_L(d)}{\partial s} s(1-s)\varepsilon_D \frac{\mathrm{d}\ln C}{\mathrm{d}\ln \alpha},$$

where the partial derivative holds resident housing requirements fixed. A population share omits differences in housing requirements across types. No numerical elasticity or calibrated affordability effect is asserted here: the supplied estimates do not identify all components on a common metropolitan basis.

In a richer accommodation market, effective tourist housing demand can fall without fewer tourist arrivals. Visitors could use hotels, shorten stays, or consume less residential housing per stay. Conversely, local listing disappearance may reflect movement to medium-term rentals, inactivity, or a change in platform coverage rather than a residential conversion. These possibilities require measurement of housing use across the metropolitan market, rather than an inference from listing counts alone.

4.2 Other adjustment margins and policy scope

Resident mobility changes the incidence of an aggregate demand reduction. If resident numbers also respond, the line-city accounting becomes

$$\mathrm{d}R_L(d) = \beta_L\big(\ell_L \,\mathrm{d}N_L + \ell_S \,\mathrm{d}N_S\big).$$

Additional residential demand can absorb housing released by visitors. With a fixed outside utility for freely mobile residents and fixed budgets and travel costs, the residential bid-rent schedule

is instead pinned by that outside option; adjustment may occur through resident numbers. An open-city extension therefore requires an explicit migration closure.

Housing construction, vacancies, conversion costs, location-specific amenities, and travel frictions are also held fixed or omitted in the benchmark. These margins can produce local rent effects even when metropolitan tourist numbers change little. The metro should consequently be understood as the set of sufficiently close substitutes relevant to the policy, with its empirical boundary assessed using access and substitution patterns. It is not an assumption that every address in an administrative metropolitan area is an equally good substitute.

Coordinated regulation across close substitutes can reduce the opportunity for geographic displacement. It may therefore produce a different effect from a restriction confined to one municipality. Its affordability consequences still depend on accommodation substitution, resident demand, and housing supply. Once a policy exhausts the capacity described in eq. (13), the fixed-demand result cannot be extrapolated to it.

4.3 Distributional effects

The renter and visitor utility results are not a complete welfare assessment. Define $K(z) = \int_0^{\max\{z,0\}} F(v)\,\mathrm{d}v$. Expected net landlord revenue per unit of land is

$$\Pi(d) = R_L(d) + \alpha(d)K(A - \kappa d). \tag{18}$$

Outside a local restriction, unchanged eligibility and a higher premium raise expected returns where short-term opportunities exist. Inside, lower eligibility and higher visitor bids have opposing effects; a complete ban removes the hosting option. Thus unchanged resident renter utility does not mean that the policy has no beneficiaries. Capitalizing these returns into sale prices requires additional assumptions about expectations, permanence, and discounting. Hotels, neighbourhood externalities, wages, tax revenue, and public spending are absent from the model, so the propositions do not establish an aggregate welfare or tourism-revenue effect.

5 Empirical Setting, Design, and Evidence

The empirical analysis asks whether the evidence has the joint pattern implied by metropolitan substitution. I distinguish reported results from the additional quantities needed to identify the mechanism's magnitude. In particular, a lack of a sustained residential rent decline, higher visitor prices, and peripheral growth are informative together, but do not uniquely identify complete displacement.

5.1 Setting and data

NYC passed Local Law 18 in January 2022 and began enforcement in September 2023. The law requires hosts to register and be present during stays, and restricts platforms from processessing unregistered bookings. As a result, the number of available short-term rentals in the city fell by approximately 90% in the first few months of enforcement. This is a statement about the reported listing measure, not a measured return of the same number of dwelling units to the long-term rental stock. The treatment date is enforcement; anticipation after enactment remains a possible source of pre-enforcement adjustment.

Following the presentation, the data sources are InsideAirbnb and AirDNA for short-term accommodation, Dwellsy and RentHub for long-term rental listings, and ATTOM for property sales. The presentation lists coverage of 2022–2025 for the short-term rental sources and 2021–2025 for the residential and sales sources. These are source-coverage windows, rather than verified estimation windows for every result. The synthetic comparison figures shown in the presentation cover a shorter period ending in 2024.

InsideAirbnb provides listing identifiers, approximate locations, listing type, unit characteristics, minimum stays, and advertised daily prices for covered locations. AirDNA extends the geographic coverage through submarket aggregates on listings, occupancy, bookings, revenue, and daily rates. The original draft describes 82 NYC submarkets and 51 surrounding submarkets; that geographic coverage should be reconciled with the updated presentation sample before treating the panel as an exhaustive metropolitan total.

Dwellsy and RentHub provide residential asking-rent observations. Repeated observations on the same property allow property fixed effects, but do not eliminate changes in which units are

advertised or observed repeatedly. The original draft limits each apartment to one observation per quarter; integration of RentHub requires a consistent period definition and removal of duplicate properties across sources. ATTOM provides recorded sales for a potential analysis of capitalization. The principal evidence below concerns rents and short-term activity, following the presentation's main results.

5.2 Residential rents: local and metropolitan comparisons

For property i in period t, an event-study specification comparing NYC with the rest of its metro is

$$\ln r_{it} = \mu_i + \lambda_t + \sum_{k \neq -1} \theta_k \big(T_i \mathbf{1}\{t - t_0 = k\} \big) + e_{it}, \tag{19}$$

where T_i indicates NYC, μ_i are property fixed effects, λ_t are period fixed effects, and t_0 is the enforcement period. Here $k = -1$ denotes the omitted pre-enforcement period; the presentation uses a different displayed normalization, which must be reconciled with the estimation code before reproducing its coefficients. The original draft also considers differential exposure within NYC, defining high-Airbnb areas using the pre-policy distribution of short-term rental activity.

The NYC-versus-metro design estimates a *relative* change. If the policy also affects the comparison area, then the treatment contrast combines NYC's response with that area's spillover response. It does not identify NYC's effect relative to no policy, or the total metropolitan effect, without additional assumptions. The high-exposure comparison likewise requires parallel counterfactual trends and can be affected by neighbourhood-specific demand and spatial spillovers.

The event-study figure does not display a sustained negative post-enforcement differential. It should not be interpreted as proof of equality: the intervals admit a range of effects, and the comparison is within a potentially affected market. The pooled estimates in table 1 further discipline the interpretation. The high-Airbnb estimate is 0.0059 with standard error 0.0058, whereas the NYC estimate is 0.0138 with standard error 0.0038 and is marked statistically significant in the presentation. Both point estimates are positive. These results do not support a rent reduction in their reported contrasts, but the second is not a statistical null and neither by itself identifies a metropolitan causal effect.

To examine the aggregate effect, the presentation also compares NYC and the New York

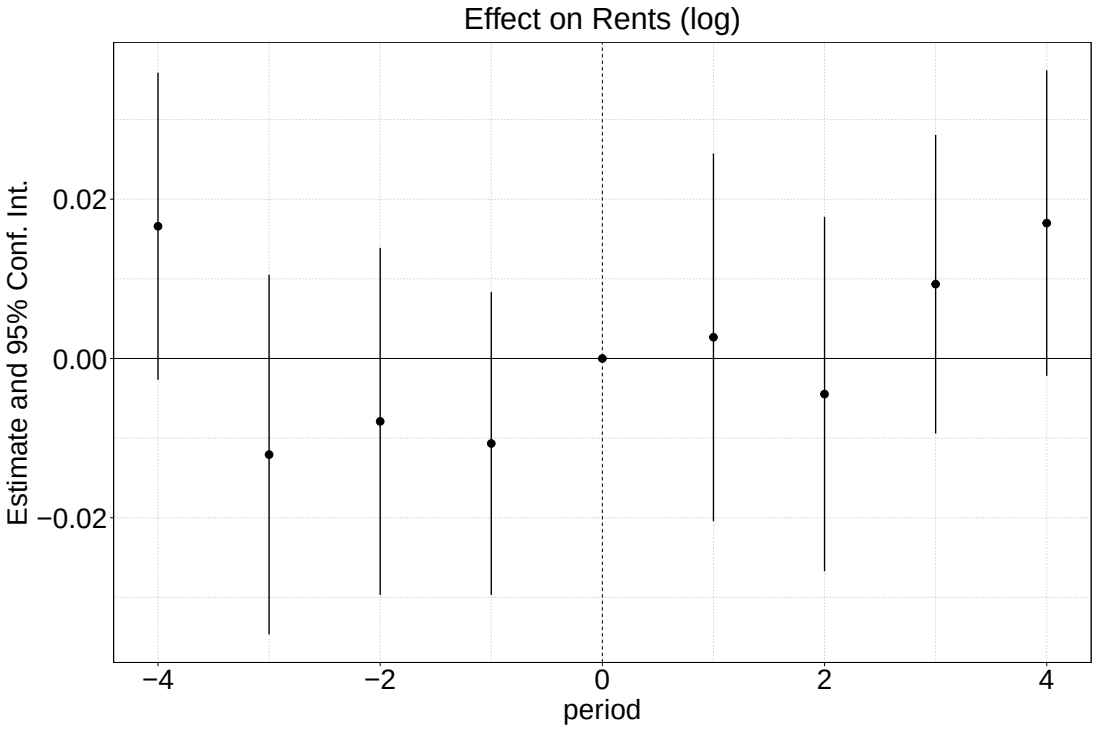


Figure 1: Residential rent event study: NYC relative to the rest of the metropolitan area. Reproduced from the CEA presentation. The plotted intervals are labelled as 95% confidence intervals. The source normalization is retained in the figure.

Table 1: Residential rent estimates reported in the CEA presentation

	High-Airbnb ZIP codes	NYC
	(1)	(2)
Post	0.0527***	0.0400***
	(0.0030)	(0.0028)
Treated × Post	0.0059	0.0138***
	(0.0058)	(0.0038)
Observations	38,414	50,398
R^2	0.9215	0.9581
Within R^2	0.0232	0.0241
Property fixed effects	Yes	Yes

Notes: Dependent variable: log rent. Coefficients, sample sizes, column headings, and significance markers are transcribed from the presentation's "LTR Rents Table" (PDF page 42). The presentation states that standard errors are clustered at the property level. It does not provide the full estimation windows, comparison pools, or a definition of the significance markers. The reported pooled specification should not be assumed identical to eq. (19); in particular, its separately estimated Post coefficient does not establish unrestricted period fixed effects.

metropolitan area with synthetic comparisons constructed from other US cities without major short-term rental restrictions during the study period. The original draft describes a synthetic difference-in-differences approach (Arkhangelsky et al. 2021). The metropolitan comparison is particularly relevant because it includes the nearby locations that may absorb displaced activity.

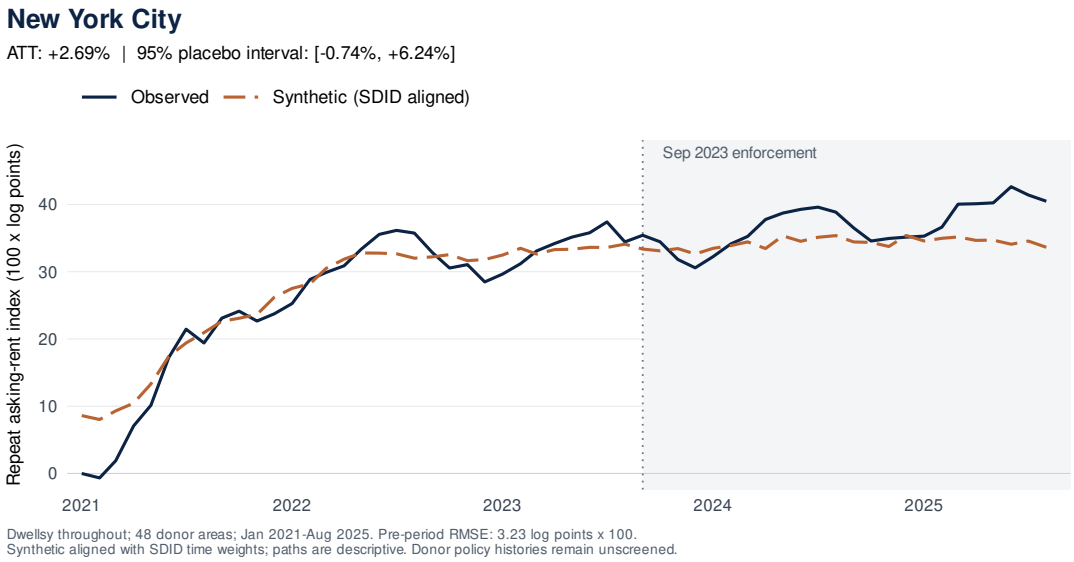


Figure 2: NYC residential rent index and its synthetic comparison. Reproduced from the CEA presentation. The plotted paths are descriptive evidence about the comparison; the figure alone does not provide the synthetic difference-in-differences estimate or its uncertainty.

Neither plotted comparison exhibits a persistent post-enforcement gap with New York rents below the synthetic series: an initial relative decline is followed by an increase. This visual pattern is consistent with limited sustained affordability gains, but the supplied figures do not report a treatment-effect estimate, confidence interval, or a smallest detectable effect. Those objects are necessary to distinguish a small effect from insufficient precision. The donor pool, geographic definitions, pre-period fit, weighting procedure, and sensitivity to individual donors also need to accompany a causal interpretation. A rising observed rent index alone does not establish that rents would not have risen faster without the policy.

Property-level clustering, as reported in the presentation, addresses serial dependence within a property but does not by itself address common policy or spatial shocks. Inference should reflect the level at which treatment and exposure vary, with the limited number of treated markets made explicit. Changes in rental-market composition, concurrent local shocks, and adjustment before

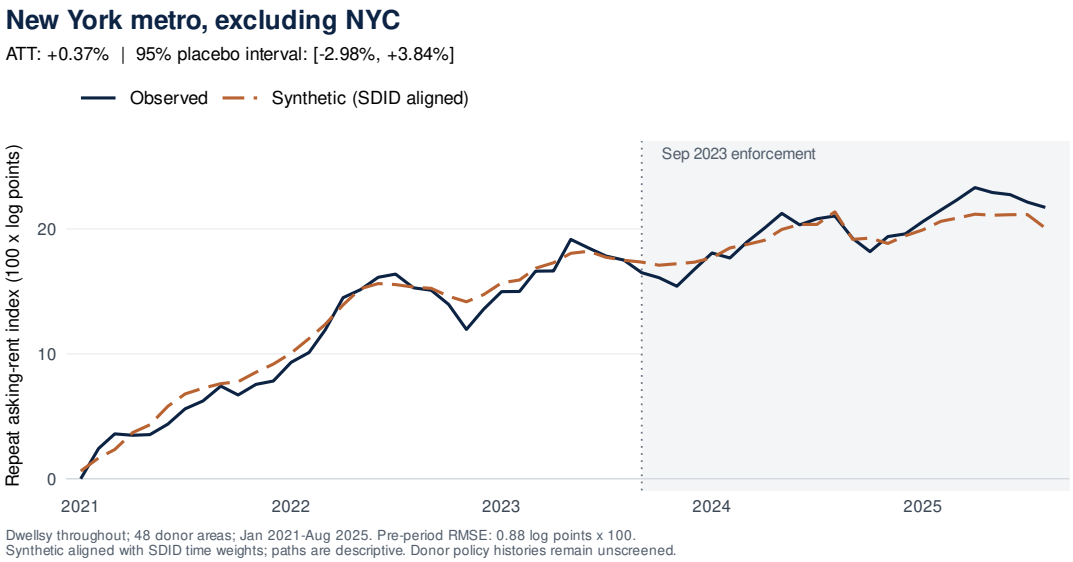


Figure 3: New York metropolitan residential rent index and its synthetic comparison. Reproduced from the CEA presentation. The metropolitan series is identified by the source legend, despite the source plot's generic NYC title.

enforcement remain potential confounders.

5.3 Short-term rental prices

Proposition 2 predicts a higher metropolitan visitor bid-rent schedule. The presentation's estimated pre- and post-period price-distance profiles show higher predicted Airbnb prices throughout the displayed distance range. This pattern is consistent with increased visitor accommodation costs. It is not, by itself, a test that the policy caused a common additive price shift.

The original draft describes hedonic regressions controlling for observable listing characteristics and a repeat-listing analysis restricted to listings that retain the same accommodation type. Such restrictions help separate price changes from the conversion of entire-home listings into rooms. They do not eliminate selection into survival, changes in service quality, or unobserved listing characteristics. A within-listing time index without an unaffected comparison also includes seasonality and other metropolitan demand shocks.

The model is stated in common-period rent per unit of housing. Advertised daily prices, realized nightly revenue, and monthly residential rents are different objects unless duration, occupancy, unit size, and included services are harmonized. If the model predicts $R_{S,1}(d) = R_{S,0}(d) + \delta$, the

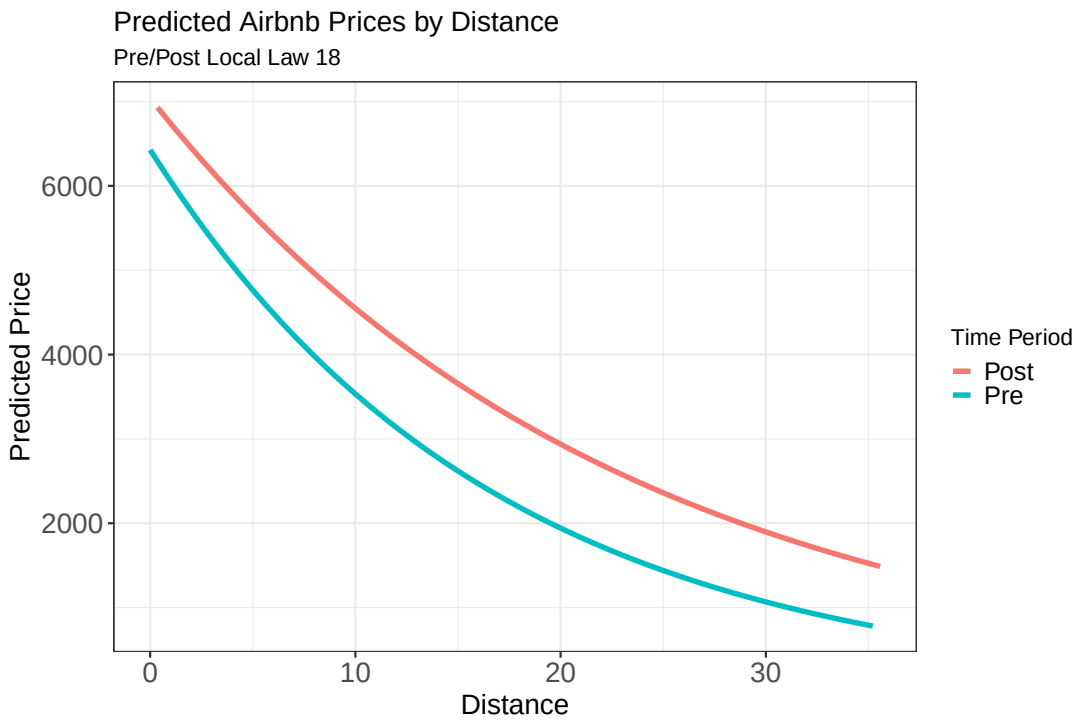


Figure 4: Predicted Airbnb prices by distance before and after enforcement. Reproduced from the CEA presentation. Price-period normalization, distance units, and estimation uncertainty should be supplied with the underlying figure before a quantitative comparison with the model.

corresponding log change is

$$\ln R_{S,1}(d) - \ln R_{S,0}(d) = \ln\left(1 + \frac{\delta}{R_{S,0}(d)}\right),$$

which varies with the initial rent. Accordingly, a constant post coefficient in a log-price regression is not the model's additive-shift restriction. The current figure supports the direction of the price prediction; its magnitude and spatial form require estimates in comparable units and an explicit joint test.

5.4 The location and amount of short-term activity

The presentation documents two geographic patterns. Within NYC, its map of changes in listing shares shows losses in central locations and relative gains in parts of the periphery. Across the broader region, the AirDNA figure shows rising indexed listings in New Jersey and Long Island alongside a decline in New York. These patterns are consistent with outward reallocation and with nearby locations serving the same destination market.

The regional series already diverge before enforcement. Anticipation, different local trends, and changing data coverage must therefore be separated from the policy response before attributing peripheral growth to Local Law 18. Moreover, the series are separately normalized. Similar percentage changes in areas with different initial listing counts do not imply offsetting levels. Establishing how much of the NYC decline was absorbed elsewhere requires common-unit counts or housing-use measures over a consistent metropolitan boundary, together with a counterfactual for each region.

It is also important to distinguish listings from bookings, nights, and housing stock. The updated presentation labels this figure as listings, whereas the original draft discussed booking growth in the corresponding part of the argument. Occupancy and duration can change independently of the number of advertised units. The model's M_S refers to housing occupied by visitors over a common period, so a quantitative test needs a transparent mapping from these measures to effective housing use.

Finally, proposition 4 predicts a supply discontinuity at a stylized policy border. The supplied map and regional time series do not estimate such a discontinuity. A border design would need comparable locations, transport access, and treatment of other regulatory differences across the

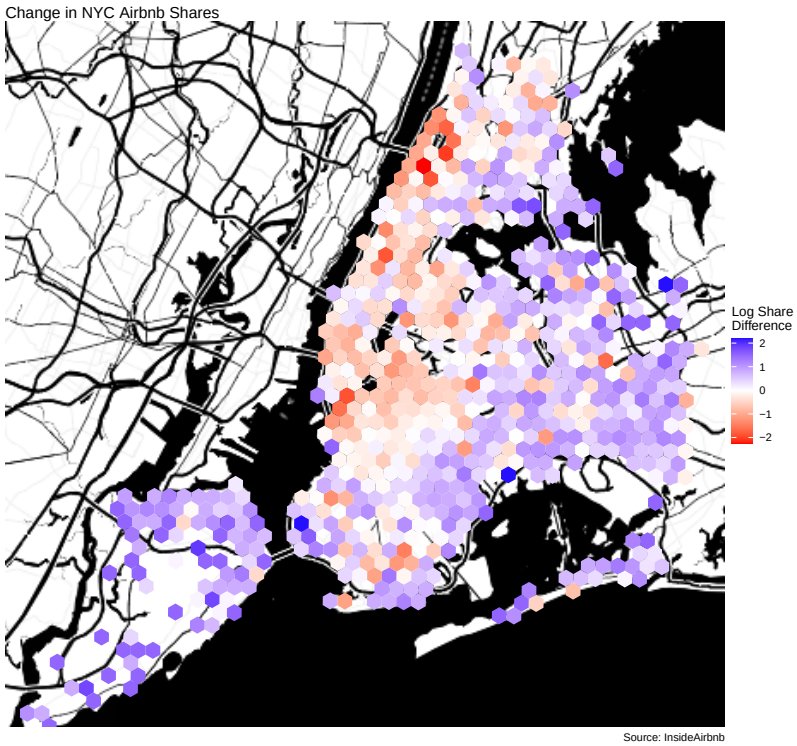


Figure 5: Changes in the spatial shares of Airbnb listings within NYC. Reproduced from the CEA presentation. The source displays log share differences. A relative share gain need not indicate growth in a location's absolute listing count.



Figure 6: Indexed short-term rental listings across the region. Reproduced from the CEA presentation's AirDNA figure: January 2023 equals one, and each series is a three-month average of total listings in the source geography. The region labels are retained from the source.

boundary; it would also face interference between nearby locations. The current evidence is evidence of spatial reallocation, rather than a completed causal border-discontinuity test.

5.5 What the joint evidence establishes

The reported evidence has the qualitative combination emphasized by the model: no sustained rent decline in the displayed comparisons, higher predicted short-term prices, and a shift of short-term activity toward peripheral locations. This combination motivates evaluating Local Law 18 at the metropolitan scale. It does not yet quantify a displacement share, establish invariant total visitor housing demand, or rule out modest affordability effects.

Conversion frictions, medium-term rental substitution, hotel substitution, and rental-sample selection may coexist with geographic displacement. The strongest empirical assessment would account for tourist housing use across the metro and estimate both municipal and metropolitan rent effects with uncertainty. Sale-price responses could provide complementary evidence on the value of hosting rights, but they need not have the same sign or magnitude as residential rent changes. The old draft's sales coefficients are not carried forward as a current result because they are absent from the presentation's main analysis and their specification is not reconciled with its updated samples.

6 Conclusion

The metropolitan market is the relevant scale for assessing whether a local short-term rental restriction expands housing availability for residents. When visitors can relocate within that market and eligible hosts elsewhere can accommodate them, a sharp contraction in one jurisdiction can be offset by expansion in another. The model makes this mechanism explicit: with fixed metropolitan housing requirements and a residential fringe, the allocation of housing changes while the residential rent schedule does not. Visitor accommodation becomes more expensive, and the distribution of short-term use shifts outward.

The evidence reported for New York City's Local Law 18 is consistent with this mechanism. The residential rent comparisons show no sustained reduction, while predicted short-term prices rise and peripheral activity expands. The empirical interpretation remains conditional on the

comparison design, the measurement of activity, and the precision of the estimates. A local collapse in listings does not measure the metropolitan increase in residential housing, and a within-metro rent comparison does not measure the total market effect.

The policy implication is to match the scale of evaluation to the scale of substitution. Local changes can have substantial effects on where activity occurs while producing limited changes in market-wide affordability when the rest of the metro can absorb them. Policies covering a broader set of substitutes may limit that offset, but their effects depend on aggregate tourist housing demand, resident mobility, housing supply, and accommodation alternatives. The central question is therefore how much a policy changes demand and availability across the relevant market, rather than how much activity disappears within its administrative boundary.

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